

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and owning distributed systems at scale. Built and operated infrastructure serving billions of daily requests across Google Search, YouTube video processing, and marketplace platforms. Led teams of 8 engineers delivering developer tooling and internal platforms consumed by thousands of engineers. End to end ownership from architecture through production incidents with on-call rotations. Partners closely with product and cross-functional teams to define technical strategy. Current focus on large-scale distributed systems and API platform reliability. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Distributed Systems: microservices, distributed systems, search infrastructure, video streaming, API platform, CDN

Infrastructure: Kubernetes, GCP, Azure, TensorFlow, BigQuery, Spanner

Languages: C++, Go, Python, Java, Ruby on Rails, TypeScript, JavaScript, HTML, CSS

Data Stores: PostgreSQL, Redis, Bigtable, BigQuery, Spanner

Observability: monitoring, on-call, incident response, SLO/SLA management

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Own video transcoding and delivery pipeline processing over a billion daily views across global CDN infrastructure.
- Collaborate with product teams to define technical requirements for new video format support and delivery optimization.
- Extended search infrastructure serving hundreds of millions of queries daily with sub-100ms p99 latency requirements.
- Designed adaptive bitrate streaming system that reduced rebuffering rates and improved viewer engagement metrics.
- Participate in on-call rotation for 99.99% availability targets across video platform services.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Led marketplace infrastructure rebuild in **Ruby on Rails** and **Go** supporting \$1B+ annual gross merchandise value.
- Redesigned payment processing pipeline reducing transaction failures by 35% through improved retry logic and idempotency.
- Owned video streaming infrastructure for live auctions handling thousands of concurrent viewers with sub-second latency.
- Established service reliability standards including SLO definitions, error budgets, and on-call runbooks across platform services.
- Mentored 4 engineers on distributed systems design patterns and Ruby on Rails backend architecture.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of 8 engineers building internal developer platform serving 25,000+ engineers across **Python**, **Go**, and **Java** codebases.
- Designed and shipped unified API gateway handling 2M+ requests per second with **99.99% availability** and sub-50ms latency.
- Led migration of legacy monolith services to **microservices** architecture in **Kubernetes**, reducing deployment times by 60%.

- Architected multi-tenant A/B testing infrastructure for search ranking experiments across desktop and mobile surfaces.
- Defined technical roadmap for developer productivity tooling, aligning with organization-wide engineering efficiency goals.

Software Engineer | Google | 02/2011 - 02/2015

- Built backend systems for search quality improvements serving 100M+ daily active users with measurable ranking gains.
- Developed **C++** and **Python** components for core ranking algorithms processing petabytes of index data daily.
- Created internal experimentation framework enabling rapid A/B testing of search result ranking changes across markets.
- Collaborated with search infrastructure teams to optimize index serving latency and throughput on **GCP**.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Built and maintained **.NET Framework** APIs consumed by thousands of internal and external developers across Microsoft products.
- Developed **C++** performance-critical components for Windows kernel-level subsystems with strict memory and latency constraints.
- Contributed to **Azure** early infrastructure development supporting cloud service deployment and management workflows.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Implemented **C++** utility library components for internal developer tooling used across multiple product teams.

Education

B.S. Honors Degree in Computer Science

Virginia Tech | 2004 - 2007